

2DEG-HEATED AlGaIn/GaN MICRO-HOTPLATES FOR HIGH-TEMPERATURE CHEMICAL SENSING MICROSYSTEMS

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ABSTRACT

Fully-suspended AlGaIn/GaN micro-hotplates that leverage self-heating of the two-dimensional electron gas (2DEG) have been modeled, micro-fabricated, and characterized at elevated ambient temperatures (from 25°C to 600°C in air). An input power of ~75 mW heated the micro-hotplates to approximately 270°C from 25°C ambient conditions. In addition, finite element analysis (FEA) showed high-temperature uniformity across the micro-hotplate (<1% variation) and fast transient response times (~2 ms rise and fall times). These results support the use of the AlGaIn/GaN-on-Si platform for high-temperature activation of chemical sensing catalysts and in-situ chemical sensing within high-temperature environments (e.g., combustion exhaust, industrial process, and downhole).

INTRODUCTION

The global market of chemical sensors is forecasted to reach about 31 billion US dollars by 2020 [1]. Data collected by chemical sensors can aid in emission control, energy savings, environmental monitoring, and catastrophe prevention [2-5]. For chemical sensing, a functional sensing layer (catalyst) transduces the chemical signals into electrical signals, which are then detected by the underlying sensor platform. To ensure reliable signals and reduced baseline drift, the sensing layer needs to be regenerated (burned off) periodically at elevated temperatures. In addition, many functional sensing materials exhibit increased sensitivity at elevated temperatures. For example, ZnO nanorods demonstrated enhanced sensitivity to hydrogen at approximately 250°C [6]. To sustain elevated temperatures during operation, a heating element is often integrated into the sensor system. Micro-hotplates, which are suspended and thermally isolated (reduced power consumption) structures with embedded heaters, are often utilized.

Commonly used micro-hotplates are composed of silicon oxide/nitride membranes and metallic heaters. Such systems suffer from reliability issues caused by electro-migration of metal atoms [7]. In addition, localized stress and voids in metal traces often lead to premature rupture of the membranes and drift of the heating characteristics [7]. In addition, micro-hotplates with metal heaters often employ complex heater geometries to achieve uniform temperature distribution [8]. Thus, micro-hotplates made from doped semiconductor materials (e.g., Si and SiC) have been used to mitigate the reliability issues caused by metal electro-migration. Additionally, as the semiconductor plate itself serves as the heater, such micro-hotplates can produce uniform temperature distributions using simple layout geometries. For high-temperature chemical sensing (> 300°C), self-heated silicon carbide (SiC) micro-hotplates have been developed, but custom substrates and advanced membrane release chemistry are required [9], which makes integration with electronics challenging.

In contrast, the self-heated AlGaIn/GaN micro-hotplates reported here utilize commercial AlGaIn/GaN-on-Si wafers and a common Si release process (XeF₂), which supports monolithic integration with GaN electronics and high-temperature operation. Monolithic GaN-based microsystems are appealing for sensing applications in harsh environments, such as combustion exhaust

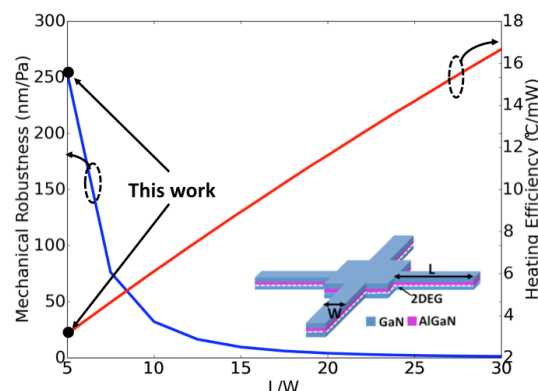


Figure 1: The influence of heater arm geometry on heating efficiency and mechanical robustness (out-of-plane displacement per unit perpendicularly applied pressure) of four-arm micro-hotplates studied by analytical modeling and FEA.

pipes, industrial manufacturing processes, and oil & gas wellbores, due to the inherent temperature- and chemical-tolerance of GaN. AlGaIn/GaN heterostructures have been commercialized for high-frequency and high-power electronics thanks to the highly conductive two-dimensional electron gas (2DEG) at the AlGaIn/GaN interface [10]. In addition, the 2DEG is sensitive to surface charges and thus can be used for chemical sensing applications [11-12]. Moreover, due to the wide bandgap of GaN, AlGaIn/GaN devices have demonstrated successful operation up to 600°C [13]. Furthermore, GaN energy harvesters and resonators [14-15] leveraging the piezoelectric properties of GaN can also be integrated on a single chip to make self-powered wireless sensing nodes. This paper presents the modeling, micro-fabrication, and characterization (interpolation-based) of the first 2DEG-heated AlGaIn/GaN micro-hotplate architecture reported in the literature. In addition, uncertainty analysis of the characterization method and future work are discussed.

DESIGN AND MODELING

Analytical modeling and finite element analysis (FEA) were used to predict the effect of micro-hotplate geometry on heating efficiency (maximum temperature rise per unit input power) and mechanical robustness (maximum out-of-plane displacement per unit perpendicularly applied pressure) (Fig. 1). The results reveal that increasing the ratio of the heater arm length (L) to width (W) can enhance the heating efficiency but reduces the mechanical robustness of the micro-hotplate. It should be noted that the mechanical robustness falls at a much faster rate than the heating efficiency improves. To ensure that the micro-hotplate has sufficient mechanical robustness, L/W = 5 was chosen for this work (L = 100 μm and W = 20 μm). For this work, a spiral four-arm layout was chosen due to the fact that given the same L/W, a spiral four-arm micro-hotplate (Fig. 2a) spans a much smaller area than a conventional four-arm design (schematic shown in the inset of Fig. 1) while maintaining a similar heating efficiency and mechanical robustness. To ensure reliable operation, the thermal stress in the spiral four-arm micro-hotplate